ST3236 CheatSheet AY25/26 — @Jinhang

Probability Foundation

Indicator Function: $\mathbb{I}[A](\omega) = \begin{cases} 1 & \text{if } \omega \in A \\ 0 & \text{if } \omega \notin A \end{cases}$

- $\mathbb{I}(A) \sim$ Bernoulli$(p_1) : \Omega \to \{0,1\}$ is a random variable
- PMF: $p_1 = P(P(A) = 1) = P(A), p_0 = P(I(A) = 0) = 1 - P(A)$
- $\mathbb{I}(A^c) = 1 - I(A) \sim$ Bernoulli(p_0)
- $\mathbb{I}(A \cap B) = \mathbb{I}(A) \times \mathbb{I}(B)$
- $\mathbb{I}(A \cup B) = \mathbb{I}(A) + \mathbb{I}(B) - \mathbb{I}(A) \times \mathbb{I}(B)$

Random Variable: also called a **Distribution**

Expectation: Center of dist. that minimizes $\mathbb{E}[(X-c)^2]$

- If $X \geq 0$ and $\mathbb{E}[X] = 0$, then $P(X = 0) = 1$
- $\mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y]$ doesn't require independence.

Variance: $\mathrm{Var}(X) = 0 \Rightarrow X$ is a constant. $P(X = E[X]) = 1$

Moment-generating Function: $M_X(t) = \mathbb{E}(e^{tX})$, function of t

- Two RV have the same m.g.f. $\Rightarrow$ same distribution
- k-th moment $\mathbb{E}[X^k] = \frac{\mathrm{d}^k}{\mathrm{d}t^k} M_X(t)|_{t=0}$
- $M_{aX+b}(t) = e^{bt} M_X(at)$

Independence: $X_1, \cdots, X_n$ are independent if for any $x_1, \cdots, x_n$ $P(X_1 = x_1, X_2 = x_2, \cdots, X_n = x_n) = \prod_{i=1}^n P(X_i = x_i)$

- $X \perp Y \not\Longleftarrow\Rightarrow$ Cov(X,Y)= 0, covariance only captures the inner product relationship.
- For independent R.V. $M_Y(t) = \prod_{i=1}^n M_{X_i}(t)$

Conditional RV: $p_{X|Y}(x|y) = \frac{P(X=x,Y=y)}{P(Y=y)}, f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)}$

- Law of Total Probability: $p_X(x) = \sum_y p_{X|Y}(x|y) p_Y(y)$
- Bayes' Formula: $p_{Y|X}(y|x) = \frac{p_{X|Y}(x|y)p_Y(y)}{\sum_y p_{X|Y}(x|y)p_Y(y)}$
- $P(X_1 = x_1, \cdots, X_n = x_n | Y = y) = \prod_{i=1}^n P(X_i = x_i | Y = y)$
- $\mathbb{E}[X|Y = y] = \sum_x x * p_{X|Y=y}(x) \Rightarrow$ different for diff. y
- Law of Total Expectation: $\mathbb{E}(X) = \mathbb{E}[\mathbb{E}(X|Y)]$
- Law of Total Var.: $\mathrm{Var}(X) = \mathbb{E}[\mathrm{Var}(X|Y)] + \mathrm{Var}(\mathbb{E}(X|Y))$
- Generally, define $h(y) = \mathbb{E}[X|Y = y] : Y(\Omega) \to \mathbb{R}$, we have RV $h(Y) = (h \circ Y) : \Omega \to \mathbb{R}$ s.t. $\mathbb{E}[X|Y] := h(Y)$

Random Sum: $Y = \sum_{i=1}^N X_i$, following prop. holds for all RV

- $\mathbb{E}(\sum_{i=1}^N X_i) = \mathbb{E}(\mathbb{E}(\sum_{i=1}^N X_i | N)) = \mathbb{E}(N\mu) = \mu \mathbb{E}(N)$
- $\mathrm{Var}(\sum_{i=1}^N X_i) = \mathbb{E}[\mathrm{Var}(\sum_{i=1}^N X_i|N)] + \mathrm{Var}(\mathbb{E}[\sum_{i=1}^N X_i|N])$ $= \mathbb{E}[N\sigma^2] + \mathrm{Var}(N\mu) = \sigma^2 \mathbb{E}[N] + \mu^2 \mathrm{Var}(N)$
- Let $\phi = \sum_{i=1}^N X_i$, $M_\phi(t) = \mathbb{E}[e^{t\phi}] = \mathbb{E}[\mathbb{E}[e^{t\phi}|N]] = \mathbb{E}[(M_X(t))^N] = \mathbb{E}[e^{N \ln M_X(t)}] = M_N(\ln[M_X(t)])$

Stochastic Processes: a series of values, consider by $(\Omega, \mathcal{F}, P)$. Collection of random variables $\{X(t,\omega) : t \in T\}$

- **Series:** a collection indexed by a parameter(time, number of games/steps, generations) $\Rightarrow$ Index set T [discrete/continuous]
- (Random) **States/Values:** Numerical/Categorical. if S is
  - Countable $\Rightarrow$ Discrete-State stochastic process
  - Finite $\Rightarrow$ Finite-State stochastic process
  - Continuum $\Rightarrow$ Real-State stochastic process
- The order is **non**-random
- **Outcome Space** Ω: $\omega \in \Omega$ is a possibility that may happen
- **Set of Events** $\mathcal{F}$: each element $A \in \mathcal{F}$ is a set $A \subset \Omega$
- **Probability Measure** P: Map $\mathcal{F} \to [0,1]$, $P(\cup A_i) = \sum P(A_i)$
- **R.V.:** $X(\omega) : \Omega \to (-\infty, \infty)$ s.t. X is measurable on $\mathcal{F}$
  - For time n, we use $X_n = X_n(\omega) = X(n,\omega)$: RV $\Omega \to S$
  - Let initial state be X_0
- Sample Path: Given an outcome ω, $X(\cdot, \omega) : T \to S$

Specification MC: Consider $\{X_n; n \in \mathcal{N}\}$

- **Time:** $T = \mathcal{N}$.
- **State:** The state space S is determined by the problem.
- **Probability measure:** Transition probability matrix $\mathbf{P}$
  - **Simple events:** For $n \in \mathcal{N}$ and $i \in S \Rightarrow P(X_n = i)$
  - **Joint events:** Based on $\{X_n = i\}_{i \in S, n \in \mathcal{N}}$, for indices $n_1, n_2, \ldots, n_k \Rightarrow P(X_{n_1} = i_1, X_{n_2} = i_2, \ldots, X_{n_k} = i_k)$. Requires **joint dist.** of $(X_{n_1}, X_{n_2}, \ldots, X_{n_k}) \Rightarrow$ $p_{i_1,\ldots,i_k}(n_1, \ldots, n_k)$. [Difficult to specify for arbitrary k].

Markov Chain: S.P. $\{X_n : n \in T\}$ with discrete-state space S and discrete-time set T satisfying the Markovian property

- **Markovian property:** Given X_n, what happened afterwards is **independent** with happened in the past.
- $P(X_{n+1} = j | X_n = i, \cdots, X_0 = i_0) = P(X_{n+1} = j | X_n = i)$
- $p_{ij}^{(n,n+m)} = p_{ij}^{(m)} = P(X_{n+m} = j | X_n = i)$ $= \sum_k P(X_{n+m} = j \mid X_n = i, X_{n+1} = k) P(X_{n+1} = k \mid X_n = i)$ $= \sum_{k \in S} P(X_{n+m} = j \mid X_{n+1} = k)\, p_{ik}^{n,n+1}$ $= \sum_{k_1,\ldots,k_{m-1} \in S} [p_{ik_1}^{n,n+1} p_{k_1,k_2}^{n+1,n+2} \cdots p_{k_{m-1},j}^{n+m-1,n+m}]$
- $p_{i_1,\ldots,i_k}^{(n_1,\ldots,n_k)} = P(X_{n_1} = i_1) \prod_{t=2}^k P(X_{n_t} = i_t | X_{t-1} = i_{t-1})$
- MC with finite states must have **at least one** recurrent class
- Aperiodic MC has period $d(i) = 1$

Transition Prob. Matrix: $\mathbf{P}^{n,n+1} = (p_{ij}^{n,n+1})$

- (i,j)-th entry is $p_{ij}^{n,n+1} = P(X_{n+1} = j | X_n = i) \in [0,1]$
- Row: conditional dist. of next state $X_{n+1}|X_n = i$, sum to 1
- Chapman-Kolmogorov Equations: $\mathbf{P}^{n,n+m+1} = \mathbf{P}^{n,n+1} * \mathbf{P}^{n+1,n+m+1} = \mathbf{P}^{n,n+m} * \mathbf{P}^{n+m,n+m+1}$

Stationary Transition Prob.: $p_{ij}^{n,n+1} = p_{ij}$, the one-step transition prob. don't change when n changes.

- For stationary MC, we only need $\mathbf{P}$ in this case.
- $\mathbf{P}^{n,n+m} = \mathbf{P}^{(m)} = \mathbf{P}^m, m = 0, 1, 2, \cdots$

Short-Term Questions: Given MC $\{X_n : n \in \mathcal{N}\}$ with state S and transition prob. matrix $\mathbf{P} = (p_{ij})$, random init. state $X_0 \sim \boldsymbol{\pi}$

- Next state: $P(X_1 = j | X_0 = i) = p_{ij} = (\mathbf{P})_{ij}$
- Dist. of n steps in the future: $P(X_n = j | X_0 = i) = (\mathbf{P}^n)_{ij}$
- $P(X_n = j) = \sum_{i \in S} P(X_n = j | X_0 = i) P(X_0 = i)$ $= \sum_{i \in S} \pi(i) \mathbf{P}_{ij}^{(n)} = (\boldsymbol{\pi}\mathbf{P}^{(n)})_j \Rightarrow X_n | X_0 = i \sim \boldsymbol{\pi}\mathbf{P}^{(n)}$
- $X_n, X_m | X_0 = i, n > m$: Joint dist. of two time points $P(X_n = j, X_m = k | X_0 = i) = (\mathbf{P}^{n-m})_{kj}(\mathbf{P}^m)_{ik}$
- Past state conditional on the future: $P(X_m = k | X_n = j, X_0 = i) = \frac{P(X_m=k, X_n=j|X_0=i)}{P(X_n=j|X_0=i)} = \frac{(\mathbf{P}^{n-m})_{kj}(\mathbf{P}^m)_{ik}}{(\mathbf{P}^n)_{ij}}$

Random Walk: Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a prob. space. $\{\xi_n\}_{n \geq 1}$ be i.i.d. random vectors with dist. $\xi_n \sim p$ on $\mathcal{Z}$. $\{X_n\}_{n \geq 0}$ is a random walk if $X_0 = x_0 \in \mathbb{Z},\ X_{n+1} = X_n + \xi_{n+1},\ n \geq 0$

- $X_n = x_0 + \sum_{i=1}^n \xi_i,\ \mathbb{E}[X_n] = x_0 + n\,\mathrm{E}[\xi_1],\ \mathrm{V}(X_n) = n\,\mathrm{V}(\xi_1)$ $M_\xi(t) = \mathbb{E}[e^{t\xi_1}] \Rightarrow M_{X_n}(t) = \mathbb{E}[e^{tX_n}] = e^{tx_0}(M_\xi(t))^n$
- R.W. are **stationary** MC $S = \mathcal{Z} = \{.., -N, .., -1, 0, 1, .., N, ..\}$

1-dim RW: $\xi_n \in \{1, -1\} \Rightarrow \mathbb{E}[\xi_n] = 2p - 1, \mathrm{Var}(\xi_n) = 4p(1-p)$

- Sym./Simple R.W: $P(\xi_n = 1) = 1/2$, $\mathbb{E}[X_n] = x_0,\ \mathrm{Var}(X_n) = n$
- Biased random walk: $P(\xi_n = 1) = p \neq 1/2$
- Lazy random walk: $\xi_n \in \{-1, 1, 0\}, P(\xi_n = 0) = r > 0$
- Random walk with reflecting boundaries: for constants $a < b$ $P(X_{n+1} = a + 1 | X_n = a) = 1,\ P(X_{n+1} = b - 1 | X_n = b) = 1$

d-dim RW: $\mathcal{Z}^d = \{(x_1, \cdots, x_d) : x_j \in \mathcal{Z}\}$. At each step, the random walker chooses one of its $2d$ nearest neighbors with prob. p_i s.t. $\sum p_i = 1$. Denote that step as ξ_i, $X_n = x_0 + \sum \xi_i$

Brownian Motion: For each step the walker moves Δx, in Δt time. Let $\Delta x, \Delta t \to 0$ and $\Delta t = c\Delta x^2$.

General Case: $X_n = X_0 + \sum_{i=1}^n \xi_i$, $(\xi_i + 1)/2 \sim$ Bern.(p), with absorbing state 0, N

- If $X_0 = 0$ then $(X_n + n)/2 \sim$ Binomial(n,p)
- $p = 1/2$, $P(X_n = 0 | X_0 = k) = 1 - \frac{k}{N}$, $\mathbb{E}[T | X_0 = k] = k(N-k)$
- $p \neq 1/2$, $u_k = P(X_n = 0 | X_0 = k) = 1 - \frac{1-(q/p)^k}{1-(q/p)^N}$ $$v_k = \mathbb{E}[T|X_0 = k] = \frac{1}{p-q}\left(N\frac{1-(q/p)^k}{1-(q/p)^N} - k\right)$$
  - $p < q$, $u_k \approx 1 - \frac{1}{(q/p)^{N-k}}$, $N \to \infty, u_k \to 1$ Broke finally. $v_k \uparrow \to \frac{-k}{p-q} = \frac{k}{q-p}$
  - $p > q$, $u_k \approx (q/p)^k - (q/p)^N$, $N \to \infty, u_k \to (q/p)^k$, $v_k \uparrow \to \frac{1}{p-q}[N(1-(q/p)^k) - k] \to \infty$
  - $k \to \infty, u_k \downarrow$. Fix k, $p \uparrow \Rightarrow u_k \downarrow$
- **Recurrent vs Transient:** (no absorbing boundary) $P_{00}^{(2k)} = \binom{2k}{k} p^k (1-p)^k \approx \frac{(4p(1-p))^k}{\sqrt{\pi k}} = \frac{(4pq)^k}{\sqrt{\pi k}}$
  - $p = \frac{1}{2} \Rightarrow 4pq = 1$, $\sum_k P_{00}^{(2k)} \sim \sum_k \frac{1}{\sqrt{k}} = \infty \Rightarrow$ **Recurrent**.
  - $p \neq \frac{1}{2} \Rightarrow 4pq < 1$, $\sum_k P_{00}^{(2k)} \lesssim \sum_{k=0}^\infty (4pq)^k = \frac{1}{(1-4pq)} < \infty \Rightarrow$ **Transient** (walk drifts to one side $\Rightarrow$ finite returns)

Strategy-Related Questions

Stopping Time: Let $\{X_n\}_{n \geq 0}$ be a MC. RV. $T : \Omega \to \{0, 1, 2, .., \infty\}$ is stopping time, if $\forall n \geq 0$ event $\{T = n\}$ depends only on $X_1, ..., X_n$

- At time n, one can decide whether to stop based only on the information available **up to** time n
- $P(X_T = a \mid X_S = i, S < T) = P_i(X_T = a)$
- $\mathbb{E}[T \mid X_S = i, S < T] = S + \mathbb{E}_i[T]$

Absorbing States: State i that $p_{ij} = 0$ for all $j \neq i \Leftrightarrow p_{ii} = 1$

1. $\lim_{n \to \infty} P(X_n = i) = P(X_T = \text{Absorbing State} | X_0 = k)$ **Example:** $P(X_T = 0 | X_0 = k) \Rightarrow$ Dist. of $\mathbf{P}^T(k, 0)$ Since T is a RV, we cannot compute $\mathbf{P}^T$ directly. Hence we have $u_k = P(X_T = A | X_0 = k) = \sum_{j \in S} P(X_T = A, X_1 = j | X_0 = k)$ $= \sum_{j \in S} P(X_T = A | X_1 = j) P(X_1 = j | X_0 = k)$ $= \sum_{j \in S} u_j \cdot P(X_1 = j | X_0 = k)$ [By First Step Analysis]
   First Step Analysis Define $\{Y_n, n = 0, 1, 2, \cdots\}$ s.t. $Y_n = X_{n+1}$
   - Therefore, $P(X_T = A | X_1 = j) = P(Y_{T-1} = A | Y_0 = j)$ [1]
   - Since $\{Y_n\}$ has the same probabilistic structure with $\{X_n\}$, let $T_Y = \min\{n; Y_n \in \{0, 4\}\}$ be the stopping time for $\{Y_n\}$. $\Rightarrow P(Y_{T_Y} = A | Y_0 = j) = P(X_T = A | X_0 = j)$ [2]
   - $P(X_T = A | X_1 = j) =^{[1]} P(Y_{T-1} = A | Y_0 = j) =^{[T_Y = T-1]} P(Y_{T_Y} = A | Y_0 = j) =^{[2]} P(X_T = A | X_0 = j)$

   For each u_i set up such equation and solve the system. $(u_A = 1)$
2. $\mathbb{E}[X_T | X_0 = k] = \sum j \times p_{k,j}^{0,T}$
3. Number of Games: $\mathbb{E}[T | X_0 = k]$ $v_k = \mathbb{E}[T|X_0 = k] = \sum_{t=0}^\infty t * P(T = t | X_0 = k)$ $= \sum_{t=0}^\infty \sum_i t * P(T = t | X_0 = k, X_1 = i) P(X_1 = i | X_0 = k)$ $= \sum_i \left[\sum_{t=0}^\infty t * P(T = t | X_0 = k, X_1 = i)\right] P(X_1 = i | X_0 = k)$ $= \sum_i \mathbb{E}[T | X_0 = k, X_1 = i] P(X_1 = i | X_0 = 3) = \sum_i (1 + v_i) p_{ki}$
   - $\mathbb{E}[T | X_0 = j, X_1 = i] = \mathbb{E}[T | X_1 = i] = \mathbb{E}[T_Y + 1 | Y_0 = i] = 1 + \mathbb{E}[T_Y | Y_0 = i] = 1(\text{Time per step}) + \mathbb{E}[T | X_0 = i]$.
   - Let $v_i = \mathbb{E}[T | X_0 = i]$, $v_k = \sum_{i=0}^4 (1 + v_i) P(X_1 = i | X_0 = k)$
   - Derive the equation system for all v_i. ($v_{\text{Absorbing}} = 0$)

Long Run Performance

Accessibility: state j is accessible **from** state i, $(i \to j)$ if $\exists m \geq 0$ s.t. $P_{ij}^{(m)} > 0$

- The accessibility is not symmetric ($i \to j \not\Rightarrow j \to i$)
- Accessibility is **not** to evaluate only one step

Communicate: If $i \to j$ and $j \to i$, then $i \leftrightarrow j$.

- Reflexivity: $i \longleftrightarrow i \Rightarrow p_{ii}^{(0)} = 1$
- Symmetry: if $i \longleftrightarrow j$, then $j \longleftrightarrow i$
- Transitivity: if $i \longleftrightarrow j$, $j \longleftrightarrow k$, then $i \longleftrightarrow k$.
  $P_{ij}^{(s)} > 0, P_{jk}^{(t)} > 0 \Rightarrow P_{ik}^{(s+t)} = \sum_{l \in S} P_{il}^{(s)} P_{lk}^{(t)} \geq P_{ij}^{(s)} P_{jk}^{(t)} > 0$

Communication Class: A set of states $\mathcal{C}$ s.t. for any $i, j \in \mathcal{C}$, there is $i \longleftrightarrow j$ **and** for any $i \in \mathcal{C}$ and $j \notin \mathcal{C}$,$i \nleftrightarrow j$.

- **well-defined:** Every state belongs to one and only one class
- Either recurrent or transient [Share property in the class]
- if $i \to j$ for $i \in \mathcal{C}_1$ and $j \in \mathcal{C}_2$, then for any $i' \in \mathcal{C}_1$ and $j' \in \mathcal{C}_2$, there is $i' \to j'$. Further, since $\mathcal{C}_1 \neq \mathcal{C}_2 \Rightarrow j \nrightarrow i$ and $j' \nrightarrow i'$
- Recurrent $\mathcal{C}_k \Rightarrow P_{ij} = 0$ for any $i \in \mathcal{C}_k$ and $j \notin \mathcal{C}_k$
- MC with finite states must have at least one recurrent class

Irreducible Chain: An MC is **irreducible** if all the states comm. with one another, i.e. **only one** comm. class in the MC.

- May have infinitely many states

Reducible: $\Leftrightarrow > 1$ classes

First Return Probability: For any state i, define the prob. that starting from state i, the **first return** to i is at the nth transition

$f_{ii}^{(n)} = P(X_1 \neq i, X_2 \neq i, \cdots, X_n = i | X_0 = i)$ where $f_{ii}^{(0)} := 0$

- $f_{ii} = \sum_{n=0}^{\infty} f_{ii}^{(n)} = \lim_{N \to \infty} \sum_{n=0}^{N} f_{ii}^{(n)} \leq 1$ (prob revisit i)

# of revisiting: Let $N_i = \sum_{n=1}^{\infty} \mathbb{I}(X_n = i)$ be # of revisits state i

- $P(N_i \geq m | X_0 = i) = f_{ii}^{m}$
- $\mathbb{E}[N_i | X_0 = i] = \sum_{k=1}^{\infty} k P(N_i = k | X_0 = i)$
  $= \sum_{k=1}^{\infty} \sum_{m=1}^{k} P(N_i = k | X_0 = i)$
  $= \sum_{m=1}^{\infty} \sum_{k=m}^{\infty} P(N_i = k | X_0 = i)$
  $= \sum_{m=1}^{\infty} P(N_i \geq m | X_0 = i) = \sum_{j=1}^{\infty} f_{ii}^{m} = \frac{f_{ii}}{1 - f_{ii}}$

Return Probability: $P_{ii}^{(n)} = P(X_n = i | X_0 = i)$ [Return i in n step].

- $P_{ii}^{(0)} = 1,\ P_{ii}^{(1)} = \mathbf{P}_{ii},\ P_{ii}^{(2)} = (\mathbf{P}^2)_{ii},\ 0 \leq P_{ii}^{(n)} \leq 1$
- Possible that $\sum_{i=0}^{n} P_{ii}^{(n)} > 1$ ($\Leftarrow A_n$ not disjoint)

First Return Prob. and Return Prob.

- $E_n = \{X_0 = i, X_1 \neq i, X_2 \neq i, \cdots, X_{n-1} \neq i, X_n = i\}$ Disjoint
- $A_n = \{X_0 = i, X_n = i\}$ Not Disjoint
- $f_{ii}^{(n)} = P(E_n | X_0 = i), P_{ii}^{(n)} = P(A_n | X_0 = i)$
- $E_n \subseteq A_n \Rightarrow f_{ii}^{(n)} \leq P_{ii}^{(n)}$
- $P(\cup_{n=1}^{\infty} E_n) = \sum_{n=1}^{\infty} P(E_n) = \sum_{n=1}^{\infty} f_{ii}^{(n)} \leq 1$

Connection:

- $P_{ii}^{(n)} = \sum_{k=1}^{n} P(\text{first arrives } i \text{ at } k | X_0 = i) P(X_n = i | X_k = i)$
  $= \sum_{k=1}^{n} P(E_k | X_0 = i) P(A_{n-k} | X_0 = i) = \sum_{k=1}^{n} f_{ii}^{(k)} P_{ii}^{(n-k)}$
  $= \sum_{k=1}^{n} f_{ii}^{(k)} P_{ii}^{(n-k)} + 0 * P_{ii}^{(n)} = \sum_{k=0}^{n} f_{ii}^{(k)} P_{ii}^{(n-k)}$
- $\mathbb{E}[N_i | X_0 = i] = \mathbb{E}[\sum_{n=1}^{\infty} \mathbb{I}\{X_n = i\} | X_0 = i] = \sum_{n=1}^{\infty} \mathbb{E}[\mathbb{I}\{X_n = i\} | X_0 = i] = \sum_{n=1}^{\infty} P(X_n = i | X_0 = i) = \sum_{n=1}^{\infty} P_{ii}^{(n)}$

Recurrent: $f_{ii} = 1$, $\mathbb{E}[N_i | X_0 = i] = \infty$

Transient: $f_{ii} < 1$, $\mathbb{E}[N_i | X_0 = i] < \infty$, $f_{ii} \uparrow \Rightarrow \mathbb{E} \uparrow$. However, **never** revisit i again in the long run! ($\sum_{n=m}^{\infty} P_{ii}^{(n)} \to 0$ when $m \to \infty$)

Long Performance Analysis (With Absorbing) $\Leftrightarrow$ Find $P_{ij}^{(n)}$

1. Figure out all the recurrent/transient classes
2. Convert recurrent class $\mathcal{C}_k$ in to a absorbing state k, where $P_{kj}^{(n)} \to 0$ for $j \nleftrightarrow k$, $P_{ik} = \sum_{j \in \mathcal{C}_k} P_{ij}$, $P_{kk} = 1$
3. For transient state i: $P_{i\mathcal{C}_k}^{(n)} = P(X_n \in \mathcal{C}_k | X_0 = j) \to u_{k|j}$
4. Construct new transit matrix and let $T = \min\{n : X_n = S_k\}$
5. Let $u_i = P(X_T = k | X_0 = i)$ and solve u_i using FSA

Period: $d(i) = \gcd\{n \geq 1, P_{ii}^{(n)} > 0\}$. If $\{n \geq 1, P_{ii}^{(n)} > 0\}$ is empty (never revisit i), then define $d(i) = 0$

- $\{n \geq 1, P_{ii}^{(n)} > 0\}$: the length of all possible paths to revisit i
- $\exists N$ s.t. $\forall n \geq N,\ 0 < k < d(i),\ P_{ii}^{(n*d(i))} > 0,\ P_{ii}^{(n*d(i)+k)} = 0$
- If $\exists m > 0$ s.t. $P_{ji}^{(m)} > 0$ then for $\forall n \geq N$, $P_{ji}^{(m+n*d(i))} > 0$
- If i and j can communicate, then $d(i) = d(j)$
- $\lim_{k \to \infty} P_{ii}^{(k)}$ exists only when $P_{ii}^{(nd)} \to 0$
- $d(i) = 1 \Rightarrow P_{ii}^{(n)} > 0 \Rightarrow$ Converge to stationary/limiting dist.

Regular Markov Chain: MC with transit prob. matirx $\mathbf{P}$ s.t. $\exists k > 0$, all the elements of $\mathbf{P}^{(k)}$ are **strictly** positive.

- $P(X_k = j | X_0 = i) > 0$ for any i, j at step $k \Rightarrow$ irreducible MC (If this MC has at least 2 states, then **no** absorbing states)
- This MC is aperiodic. [Since $\pi^{(k)} = \pi^{(0)} \mathbf{P}^k$, $\pi^{(k+1)} = \pi^{(k)} \mathbf{P}$ all states have non-zero prob $\Rightarrow \gcd(k, k+1) = 1$]
- $\text{Regular} \Rightarrow$ Irreducible + Aperiodic/I+A+Finite Step$\Rightarrow$R
- Finite State is sufficient condition! (Cont. example: R.W)

Main Theorem on Regular MC: Given $S = \{1, 2, .., N\}$ and $\mathbf{P}$

- $\lim_{n \to \infty} P_{ij}^{(n)} = c_{ij} = \pi_j$ exists and does not depend on i.
- **Limiting distribution:** $\sum_{k=1}^{N} \pi_k = 1$, π is a **unique** dist on S
- $\pi = (\pi_1, \pi_2, \cdots, \pi_N)$ is **also** a **stationary** dist. of this MC
  $$\begin{cases} \pi_j = \sum_{k=1}^{N} \pi_k P_{kj},\ j = 1, 2, 3, \cdots, N, \\ \sum_{k=1}^{N} \pi_k = 1. \end{cases} \Rightarrow \pi \mathbf{P} = \pi$$
- $X_n \sim \pi^{(0)} \mathbf{P}^{(n)} \to \pi$ [initial dist. does not matter in long run]
- $n \to \infty, P(X_n = j) = \sum_{i \in S} P(X_n = j | X_0 = i) P(X_0 = i) \to \pi_j$
- Fixing a path, for any time point n
  - The proportion of visits at state j is $\frac{1}{n} \sum_{k=0}^{n-1} \mathbb{I}(X_k = j)$
  - Expected Proportion: $\mathbb{E}(\frac{1}{n} \sum_{k=0}^{n-1} \mathbb{I}(X_k = j) | X_0 = i) = \frac{1}{n} \sum_{k=0}^{n-1} P[X_k = j | X_0 = i)] \to \pi_j \Rightarrow n\pi_j$ times in long run

Calculation

1. Calculate $\mathbf{P}^n$ and let $n \to \infty$
2. Solve $\pi_j = \sum_{k=1}^{N} P_{kj} \pi_k$ & $\sum_{j=1}^{N} \pi_j = 1, j = 1, 2, \cdots, N$

General Limit Theorem: For a **recurrent irreducible** MC,

- First Return Time: $R_i = \min\{n \geq 1, X_n = i | X_0 = i\}$
- Mean duration revisits $m_i = \mathbb{E}[R_i | X_0 = i] = \sum_{n=1}^{\infty} n f_{ii}^{(n)}$
- Long-run proportion $i, j \in S$: $\lim_{n \to \infty} \frac{1}{n} \sum_{k=1}^{n} P_{ij}^{(k)} = \frac{1}{m_j}$
  - $d = 1 \Rightarrow \lim_{n \to \infty} P_{ij}^{(n)} = \pi_j = \frac{1}{m_j}$
  - $d > 1 \Rightarrow \lim_{n \to \infty} P_{jj}^{(n)} = 1/m_j$ (Only holds for diag. entry)

Null Recurrent MC: when $m_j = \infty$ (e.g. Symmetric R.W)

Positive Recurrent MC: when $m_j < \infty$

- $d > 1$ only some steps $p > 0$, and it will depend on the initial state
- $d = 1$ the limiting dist. exists, as $1/m_j$
- Finite State: irreducible + aperiodic $\Leftrightarrow$ Positive Recurrent

Ergodic MC: Positive recurrent, Irreducible, and Aperiodic MC

- For any $i, j \in S$, $\lim_{n \to \infty} P_{ij}^{(n)} = \lim_{n \to \infty} P_{jj}^{(n)} = \frac{1}{m_j}$ exists
- $\pi_j = \frac{1}{m_j}$ for equation system $\pi = \pi \mathbf{P} \Rightarrow m_j = 1/\pi_j$

Reducible Chains:

- $u_{\mathcal{C}_k} | j$: Long-run, prob. of entering any recurrent class
- If $d > 1$, we have long-run proportion at each state $1/m_j$
- If $d = 1$ with finite states, limiting dist. $\pi = \pi \mathbf{P}$.
  Long run distribution $u_{\mathcal{C}_k | i} * \pi_j | \mathcal{C}_k$
- If $d = 1$ with infinite states, the prob. at each state are
  - 0 for null recurrent
  - $\pi = \pi \mathbf{P}$ for positive recurrent

  Long run distribution $u_{\mathcal{C}_k | i} * (1/m_{j|\mathcal{C}_k})$

General Procedure

1. Each recurrent class $\mathcal{C}_k$ donate as node k. Find $u_{k|i}$ by FSA
2. For each recurrent class
   - $d = 1$, solve $\pi = \pi \mathbf{P}$ and get $\pi_{j|k} \Rightarrow \pi_{j|i} = \pi_{j|k} u_{k|i}$
   - $d > 1$ consider long-run proportion of time in each state
3. Consider init. dist. $X_0 \sim \pi^{(0)}$, $\pi_{j|\pi^{(0)}} = \sum_{i \in S} \pi_{j|i} \pi_i^{(0)}$

Global Balance Equations: prob. in one state equals to the prob. that the walker will arrive at this state from other states in one step

$\pi_j = \sum_{k \in S} \pi_k P_{kj} = \sum_{k \in S} P(X_n = k) P(X_{n+1} = j | X_n = k)$

- Focus on only **one** step
- Ergodic: the dist. converge to π
- Non-ergodic: Stationary Distribution. Given $X_0 \sim \pi \Rightarrow X_n \sim \pi$

Stationary Distribution: If $P(X_n = i) = P(X_{n+1} = i) = p_i$

- If the dist. of X_0 is not π, we cannot claim any results

	Limiting Dist.	**Stationary Dist.**
Initial dist.	$X_0 \sim \pi^{(0)}$	$X_0 \sim \pi$
First step	$X_1 \sim \pi^{(0)} \mathbf{P}$	$X_1 \sim \pi \mathbf{P} = \pi$
$n \to \infty$	$X_n \sim \pi$	$X_n \sim \pi$
Calculation	$\pi = \pi \mathbf{P}$ or $\lim_{n \to \infty} \mathbf{P}^n$	$\pi = \pi \mathbf{P}$
Uniqueness	Yes $\pi_1 = \pi_2$	No (may many solutions) it's possible $\pi_1 \neq \pi_2$
MC	Ergodic	No requirement

Connections: If the limiting dist.

- exists, limiting dist. is also a stationary dist. there is only one stationary dist., which is the limiting dist.
- not exist $\Rightarrow$ no limiting dist. Stationary dist. may not be unique.

Local Balance Equations: $\pi_i P_{ij} = \pi_j P_{ji} \Rightarrow \pi$ satisfies L.B.E

- Stronger than global. Easier to verify when it holds.
- If there exists π, so that the L.B.E are satisfied, then $\pi = \pi \mathbf{P}$
- For ergodic MC, $\exists$ sol. for G.B.E, but may not $\exists$ for L.B.E

Number of equations: Global N, Local C_N^2

Page Ranking: Consider a series of webpages S.

- Construct a Random Walk (stationary MC) with $P_{ij} = \frac{1}{\text{\# of hyperlinks on page } i}$
- Let $X_0 \sim \pi^{(0)}$, after N steps $\pi^{(n)} = \pi^{(0)}\mathbf{P}^N$.
- If MC is regular, $\pi = \lim_{N\to\infty} \pi^{(0)}\mathbf{P}^N \Rightarrow$ Order $\pi_1 > \pi_2 > \cdots > \pi_{|S|}$
- Updated Algorithm: $\tilde{\mathbf{P}} = \lambda\mathbf{P} + (1-\lambda)\pi_0\mathbf{1}^T$ with damping factor λ
- $\tilde{\mathbf{P}}$ is regular with limiting distribution $\pi^{(N)}$
- Equivalently, for each step, set $\pi_{n+1} = \lambda\pi_n\mathbf{P} + (1-\lambda)\pi_0$

Markov Chain Monte Carlo Setting up

1. Consider **irreducible, aperiodic** conditional dist. $\mathbf{Q}$ s.t. $\mathbf{Q}_{kj} = P(Y_{n+1} = j | Y_n = k)$
2. Modify $\mathbf{Q}$ s.t. $\mathbf{P} = g(\mathbf{Q}) \Rightarrow p = p\mathbf{P}$ where $p_j = \sum_{k\in S} p_k g(Q_{kj})$
   2.1 Suppose $p_i Q_{ij} \neq p_j Q_{ji}$, let $p_{\min} = \min\{Q_{ij},\ Q_{ji}\}$
   2.2 Modify $p_i Q_{ij}\alpha(i,j) = p_{\min} = p_j Q_{ji}\alpha(i,j)$ s.t. $i \leftrightarrow j$, where
   $$\alpha = \frac{p_{\min}}{p_i Q_{ij}} = \frac{\min\{p_i Q_{ij}, p_j Q_{ji}\}}{p_i Q_{ij}} = \min\left(\frac{p_j Q_{ji}}{p_i Q_{ij}}, 1\right)$$
   2.3 With $\mathbf{Q}$ and α the transition prob. $\mathbf{P}$ is
   $P_{ij} = Q_{ij}\alpha(i,j),\ P_{ii} = Q_{ii} + \sum_{k\neq i} Q_{ik}(1-\alpha(i,k))$
3. Generate a path based on $\mathbf{P}$

Hastings-Metropolis Algorithm

- In general, we want to choose Q so that it is easy to generate; and
- prob. of jumping is large ($\alpha(i,j)$ is large in general)
- Convergence is faster

Discrete: Consider $p = c\vec{b}$, where c is normalising constant so that $\sum p = 1$ and **kernel function** $\vec{b}$ is vector on S, then
$$\alpha(i,j) = \min\left(\frac{\pi_j}{\pi_i}\frac{Q_{ji}}{Q_{ij}}, 1\right) = \min\left(\frac{b_j}{b_i}\frac{Q_{ji}}{Q_{ij}}, 1\right)$$

1. Let M be the number of steps to run, X be a vector with length M.
2. Choose an irreducible Markov transition probability matrix $\mathbf{Q}$ with transition prob. $Q_{ij},\ i,j \in S$
3. Init. $n = 0,\ X_0 = k,\ k \in S$ is pre-selected.
4. while $n < M$:
   (a) Generate $Y \sim q$ where $q_j = Q_{X_n,j}$
   (b) Generate $U \sim$ Unif$(0,1)$
   (c) $U < \alpha = \min\left(\frac{b_Y Q_{Y,X_n}}{b_{X_n} Q_{X_n,Y}}, 1\right)$? $X_{n+1} = Y : X_{n+1} = X_n$
   (d) $n = n+1$
5. Return $Z = X_M$

Continuous: Sample $X \sim f$ where $f = c_N f_1$

1. Set up the number of steps N. Set X to be a vector with length N.
2. Set up one-step density functions $g_x(y) = g(y|x)$ for the transitions.
3. Init: $n = 0$, $X_0 = k \in S$
4. while $n < M$:
   (a) Generate $Y \sim g_{X_n}$, $U \sim$ Unif$(0,1)$
   (b) $U < \alpha = \min\left(\frac{f_1(Y) g_Y(X_n)}{f_1(X_n) g_{X_n}(Y)}, 1\right)$? $X_{n+1} = Y : X_{n+} = X_n$
   (c) $n = n+1$

Branching Process

Consider $\{X_n\}_{n=0}^{\infty}$, where $\{X_n\}$ is a stochastic process

- Initially X_0 individuals, after n generation X_n **independently** give rise to numbers of offspring $\xi_1^{(n)}, \xi_2^{(n)}, \cdots, \xi_{X_n}^{(n)}$, $\xi^{(n)}$ are iid RV, $P(\xi = k) = p_k$
- To identify this process, we only need the distribution of ξ
- $X_{n+1} = \xi_1^{(n)} + \xi_2^{(n)} + \cdots + \xi_{X_n}^{(n)} \Rightarrow$ Markov Chain
- **Stationary:** $P(X_{n+1} = j | X_n = i) = P(\sum_{k=0}^{i} \xi_k = j | X_n = i)$ dosen't depend on n

Probability Generating Function: For a discrete RV X, the PGF is defined as
$$\phi_X(t) = E[t^X] = \sum_{k=0}^{\infty} P(X=k)t^k$$

- Function about t characterize RV X, without any randomness
- $P(X=0) = \phi_X(0),\ P(X=k) = \frac{1}{k!}\frac{d^k}{dx^k}\phi_X(t)|_{t=0}, k = 1, 2, \cdots$
- $X \perp Y \Rightarrow \phi_{X+Y}(t) = \phi_X(t)\phi_Y(t)$
- $\mathbb{E}(X) = \phi_X'(1) = \sum_{k=0}^{\infty} k \cdot P(X=k)\cdot 1^k$
- $\phi_X''(1) = \sum_{k=0}^{\infty} k(k-1)\cdot P(X=k) = \mathbb{E}(X^2) - \mathbb{E}(X)$
- $\mathrm{Var}(X) = (\phi_X''(1) + \phi_X'(1)) - \left[\phi_X'(1)\right]^2$
- Moment Gen. $M_X(s) = \mathbb{E}(e^{sX}) = \phi_X(e^s)$

Consider $X_0 = 1$

- $\phi_{X_{n+1}|X_n}(t) = \phi_\xi^{X_n}(t)$
- $\phi_{X_{n+1}}(t) = \mathbb{E}[\mathbb{E}[\prod_{i=1}^{X_n} t^{\xi_i} | X_n]] = \mathbb{E}[\phi_\xi^{X_n}(t)] = \phi_{X_n}(\phi_\xi(t))$
- $\phi_{X_n}(t) = \phi_{X_{n-1}}(\phi_\xi(t)) = \phi_{X_{n-2}}(\phi_\xi(\phi_\xi(t))) = \cdots = \phi_\xi^{(n)}(t)$

Generally $X_0 = k$, offspring $\xi \sim \phi_\xi(t)$ s.t. $\mathbb{E}[\xi] = \mu$, Var$(\xi) = \sigma^2$

- $\phi_{X_n}(t) = [\phi_\xi^{(n)}(t)]^k$.
- $$\mathbb{E}[X_n] = k\mathbb{E}\left[\sum_{i=1}^{X_{n-1}} \xi_i^{(n-1)}\right] = k\mathbb{E}[X_{n-1}]\mathbb{E}[\xi_i^{(n-1)}]$$
  $$= k\mu\mathbb{E}[X_{n-1}] = \cdots = k\mu^n$$
- $\mathbb{E}(X_n|X_m) = X_m\mu^{n-m}$
- $$\mathbb{E}[X_m X_n] = \mathbb{E}[X_m\mathbb{E}[X_n \mid X_m]] = \mu^{n-m}\mathbb{E}[X_m^2] \quad [t]$$
  $$= \mu^{n-m}(\mathrm{Var}(X_m) + (\mathbb{E}[X_m])^2)$$
  $$= \mu^{n-m}\left[k\sigma^2\mu^{m-1}\frac{\mu^m - 1}{\mu - 1} + k^2\mu^{2m}\right]$$
  $$= k\sigma^2\mu^{n-1}\frac{\mu^m-1}{\mu-1} + k^2\mu^{n+m}$$
- $$\mathrm{Var}[X_n|X_0 = k] = k\mathrm{Var}[\sum_{i=1}^{X_n}\xi_i^{(n)}]$$
  $$= k\left[\mathrm{Var}(\mathbb{E}[S_{n-1}|X_{n-1}])) + \mathbb{E}\left[\mathrm{Var}(S_{n-1}|X_{n-1})\right]\right]$$
  $$= k\left[\mathrm{Var}(X_{n-1})(\mathbb{E}[\xi])^2 + \mathbb{E}[X_{n-1}]\mathrm{Var}(\xi)\right]$$
  $$= k\left[\mu^2\mathrm{Var}(X_{n-1}) + \sigma^2\mathbb{E}[X_{n-1}]\right]$$
  $$= k\mu^{n-1}\sigma^2\begin{cases}\frac{1-\mu^n}{1-\mu}, & \mu \neq 1\\ n, & \mu = 1\end{cases}$$

Long Run: Let $Z = \sum_{n=0}^{\infty} X_n$ (total # individuals) with $X_0 = k$

- $$\mathbb{E}[Z] = \mathbb{E}\left[\sum_{i=1}^{k} Z_i^{(1)}\right] = \sum_{i=1}^{k}\mathbb{E}[Z_i^{(1)}] = \begin{cases}\frac{k}{1-\mu} & \mu < 1\\ \infty & \mu \geq 1\end{cases}$$

Extinction Probability: let $T = \min\{n, X_0 = 0\}$

- 0 is an absorbing state
- $u_n^{(k)} = P(X_n = 0|X_0 = k) = P(T \leq n | X_0 = k) = \sum_{l\in S} p_{lk}u_{n-1}^{(l)}$
  $= \prod_{i=1}^{k} P(T_i \leq n | X_0 = 1) = (u_n^{(1)})^k = u_n^k$
- $u_n = \sum_{l\in S} p_{l1}u_{n-1}^{(l)} = \sum_{l\in S} P(\xi = l)u_{n-1}^l = \phi_\xi(u_{n-1})$
- Algorithm:
  1. Set $u_1 = P(\xi = 0)$
  2. Iterating $u_m = \phi_\xi(u_{m-1})$ for $m = 2, 3, \ldots$ to get u_n
  3. $u_n^{(k)} = u_n^k$
- Prob. the population eventually dies out
  $$u_\infty^{(k)} = \lim_{n\to\infty} u_n^{(k)} = \lim_{n\to\infty} u_n^k \Rightarrow u_\infty^{(k)} = u_\infty^k$$
  $$\lim_{n\to\infty} u_n = \lim_{n\to\infty}\phi_\xi(u_{n-1}) \Rightarrow u_\infty = \phi_\xi(u_\infty)$$

  u_∞ must be solution of $x = \phi_\xi(x),\ x \in [0,1]$
  - $P(\xi = 0) = 0 \Rightarrow u_1 = 0, \phi(0) = P(\xi = 0) = 0 \Rightarrow u_\infty = 0$
  - If $P(\xi = 0) > 0$, consider
    $\frac{\mathrm{d}}{\mathrm{d}x}\phi_\xi(x) = \sum_{k=1}^{\infty} P(\xi = k)\cdot k \cdot x^{k-1} > 0$
    $\frac{\mathrm{d}^2}{\mathrm{d}x^2}\phi_\xi(x) = \sum_{k=2}^{\infty} P(\xi = k)\cdot k(k-1)\cdot x^{k-2} \geq 0$
    $\frac{\mathrm{d}}{\mathrm{d}x}\phi_\xi(x)|_{x=1} = \sum_{k=1}^{\infty} P(\xi = k)\cdot k \cdot 1^{k-1} = \mathbb{E}[\xi]$
    * $E[\xi] < 1$: subcritical, $u_\infty = 1$
    * $E[\xi] = 1$: critical, $u_\infty = 1$
    * $E[\xi] > 1$: supercritical, $u_\infty < 1$, solution of $x = \phi_\xi(x)$

Poisson Distribution: $X \sim$Pois(λ), p.m.f $p(x) = \frac{\lambda^x}{x!}e^{-\lambda}$

- Number of events on $(0, t_0]$
- $\mathbb{E}(X) = \lambda,\ \mathrm{Var}(X) = \lambda$, p.g.f. $f(t) = e^{\lambda(t-1)}$
- $X \sim$Pois(λ), $Y \sim$Pois(μ), $X + Y \sim$Pois$(\lambda + \mu)$

Poisson Process: continuous time, discrete state, with rate/intensity $\lambda > 0$

- Setup: consider $n \to \infty$, #Events$\sim$ Bin(n, p)
  $$P(X = x) = \binom{n}{x} p_n^x(1-p_n)^{n-x} \approx \frac{(\frac{n}{e})^n}{x!(\frac{n-x}{e})^{n-x}} p_n^x(1-p_n)^{n-x}$$
  $$\approx \frac{(np_n)^x}{x!}e^{-x}(1 + \frac{x - np_n}{n-x})^{n-x} \approx \frac{\lambda^x}{x!}e^{-\lambda}$$
  Therefore $n \to \infty,\ p_n \to 0$, when $np_n \to \lambda$, Bin$(n, p_n) \to$ Poisson(np_n)
- $X(t)$: # of interest until time t for $t > 0$, $X(0) = 0$, $X(t_n) \sim$Pois(λt_n)
- $\mathbb{E}[X(t)] = \lambda t,\ \mathrm{Var}[X(t)] = \lambda t$
- Increments $X(t_{i+1}) - X(t_i)$ are **independent** for $t \geq 0$
- For $s \geq 0$ and $t > 0$, RV $X(s+t) - X(s) \sim$ Poisson(λt)

Rare Events: on a small interval with length h, the prob. of two events is $o(h)$

- $h \to 0$, $o(h)$ means any sequence that goes to 0 faster than h ($\lim_{h\to 0} o(h)/h = 0$)
- For each small time interval $(t, t+h)$
  - the event doesn't happen with a prob. $1 - \lambda h - o(h)$
  - the event happens once with a prob. λh
  - the events happen twice or more with prob. $o(h)$
- Non-homogeneous Poisson process: $\lambda \to \lambda(t)$
- Poisson point process: $h \to Vol(A)$

The Law of Rare Event: Let RV $N((s,t])$ be # of events in $(s,t]$. Then $N((s,t])$ is a Poisson process of intensity $\lambda > 0$ if

1. $N((t_{i+1}, t_i])$ are **independent** for $t \geq 0$
2. $\exists\lambda > 0$ s.t. $P(N((t, t+h]) \geq 1) = \lambda h + o(h)$ when $h \to 0$
3. $P(N((t, t+h]) \geq 2) = o(h),\ h \to 0$

Waiting Time: W_n, time of occurrence of the n-th event, $W_0 = 0$

- $\{W_n \leq t\} = \{$At least n event happens before time $t\} = \{X(t) \geq n\}$
  $P(W_1 \leq t) = P(X(t) \geq 1) = 1 - P(X(t) = 0) = 1 - e^{-\lambda t}$
- $f_{W_1}(t) = \frac{\mathrm{d}}{\mathrm{d}t}P(W_1 \leq t) = \lambda e^{-\lambda t} \Rightarrow W_1 \sim$Exp$(\lambda)$
- $f_{W_n}(t) = \frac{\lambda^n t^{n-1}}{(n-1)!}e^{-\lambda t},\ n = 1, 2, 3, \cdots \Rightarrow W_n \sim$ Gamma.
- **Memorylessness:** $X \sim$ Exp. $\Rightarrow P(X > t + s | X > s) = P(X > t)$
- Consider $X(t) = n$ the joint dist. of $W_1, W_2, \cdots, W_n$
  $f(w_1, w_2, \cdots, w_n | X(t) = n) = \frac{n!}{t^n}$ (n indep. Unif$(0,t)$ rv)

Sojourn times: $S_n = W_{n+1} - W_n$, time between two occurrences, measures the duration that Poisson process sojourns in state n

- $f_{S_i}(t) = \lambda e^{-\lambda t}, t \geq 0 \Rightarrow S_i \sim$ Exp(λ).
- For iid RV $\{S_n; n \geq 0\}$, $W_n = S_0 + S_1 + \cdots + S_{n-1}$ is a PP with λ

Arriving Times: Given that $X(t) = n$, the n arrival/waiting times $W_1, W_2, \cdots, W_n$ have the same distribution as the order statistics corresponding to n independent random variables uniformly distributed on the interval $(0,t)$, which is $f_k(x) = \frac{n!}{(n-k)!(k-1)!}\frac{1}{t}(\frac{x}{t})^{k-1}[\frac{t-x}{t}]^{n-k}$

A1 Series Number Convergence

- When $|r| < 1$, $\sum_{n=0}^{\infty} ar^n = \sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}$
- $\sum_{n=1}^{\infty} ar^n = \frac{ar}{1-r}$
- $\sum_{n=1}^{\infty} np^{n-1}(1-p) = 1/(1-p)$

A2 MISC

- Symmetric R.W on odd circle $d(i) = 1$, even circle $d(i) = 2$